DV2618/DV2619 – Assignment 2 – A* Search algorithm, Genetic Algorithm, Data Science and Implementations

This is individual assessed assignment. You are allowed to discuss this assessment with other students, but you should not copy their code, and you should not share your own code with other students. Note that we will carry out plagiarism checks on all submissions.

This assignment has **3 different tasks** and requires you to apply your knowledge of Genetic Algorithms, Graph Search, Data science and implementation (**preferably in Python)**. These topics are covered in the learning materials in Lectures 4, 5, 6 and 7 and it is strongly recommended that you work through the presentations from these weeks.

The **deadline for completing this assignment is the 8th October 2025, 23:59 pm**. You should submit your work (**e.g., zip file including your report, implementation etc.**) on Canvas course webpage. Note that, the zip file should be named with your name and surname (e.g., name_surname.zip).

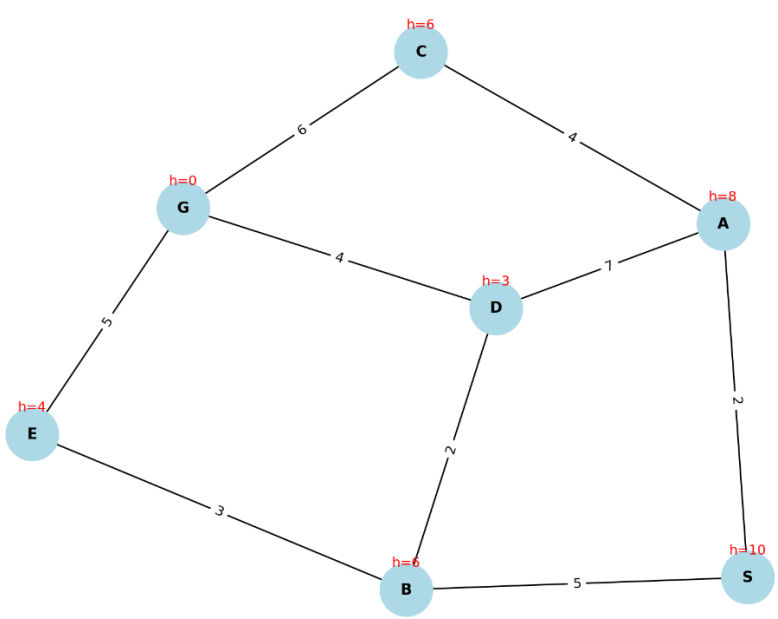
- **Reading advice:** Russell & Norvig: Artificial Intelligence - A Modern Approach, Section 4.3 in the 2nd edition.

Tasks

Taks 1 (25 Points):

A delivery robot must travel between locations in a country. The map is represented as a weighted graph where nodes are cities and edges represent travel costs (in kilometers). The goal is that you will design an agent using A* search algorithm to find the cheapest path from Start (S) to Goal (G).

The graph is shown below.
Edge Costs are in blue.
Heuristic Values (estimated distance to Goal G) are given in red above each node in the graph.



Fugure 1. Cities in a country.

a) **Apply the A* search algorithm to find the cheapest path from S to G.**
b) **In the report, show your step-by-step work:**
   - **The f(n) = g(n) + h(n) values for nodes as they are expanded.**
   - **Identify the final path found by A*.**
   - **State the total cost of the solution path.**

* For Task 1, you don't need to submit your implementation.

Tasks 2 (45 Points):

Problem Definition: You are going to solve a maze problem using GA. The aim of this task is to find a path in a maze, shown in Figure 2, for a mouse to reach the destination or food with the shortest and acceptable path. You are going to implement a Genetic Algorithm that will find a solution to find a path by optimizing the problem automatically.

- **In this task, the implementation should be done in Python, and you should submit the *implementation in .py or ipynb*.**
- **In your report, you should discuss the results that you obtained and explain how the Genetic Algorithm (GA) reaches the final solution. You should also describe the parameters you used in the algorithm, such as population size, crossover rate, mutation rate, and number of generations, and provide clear reasoning for why you chose those values. Finally, reflect on how these parameter choices influenced the outcome of your results.**

Hint: To solve this problem, you need to create a 2D - matrix where the possible passes should be 1 and obstacles must be a very large value (e.g., 1000). The matrix should be of the same size as the maze matrix.

More information and details about the problem can be found on the following webpage:
https://www.educative.io/answers/what-is-the-maze-problem

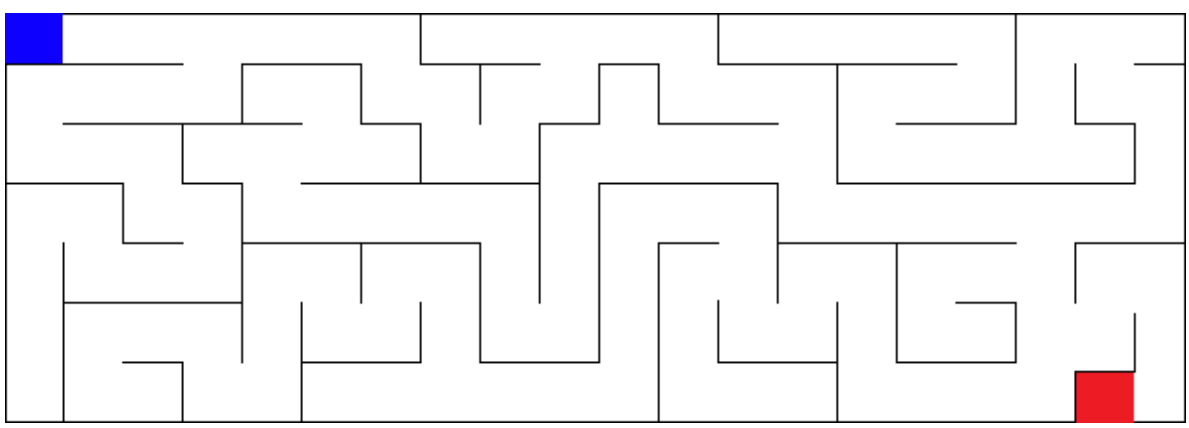


Figure 2. Maze.

Tasks 3 (30 Points):

Use Python software to complete this assignment. You can use sklearn, keras and/or tensorlow to complete the assignment.

1. Regression for Data

1.1 Generate data points in Python using:

x_data = np.linspace(−0.5, 0.5, 200)[:, np.newaxis]

```
noise = np.random.normal(0, 0.02, x_data.shape)
y_data = np.square(x_data) + noise
```

a. Establish a linear regression model to predict y data (you can use sklearn).
b. Establish a polynomial model with highest power of 2 to predict y data (you can use sklearn).
c. Establish a three-layer (including input, 1 hidden (6 nodes) and output layer) neural network to predict y data (keras or tensorflow is recommended). Split the dataset into training 80% and testing data 20% using the following code in Python:

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x_data, y_data, test_size = 0.2, random_state = 0)
```

d. Calculate and compare mean squared errors of three models in 1.1 (a), (b) and (c).
e. Plot data points and curve of predictions in 1.1 (a) and 1.1 (b).

*** In the report, briefly explain what you have done in (a), (b), (c), (d), and (e). Also, you must submit your whole python code with the report.**

Evaluation

The assessment criteria document for the Assignment 2 is provided below.

Total Mark Range	Grade
>85	A
75-84	B
65-74	C
55-64	D
40-54	E
0-39	F

Important Note: Please ensure that your code is fully functional and does not require any debugging or modifications. Teachers will not spend additional effort troubleshooting or running your code. For full consideration and a successful evaluation of your assignment, your code must work seamlessly as provided. Additionally, you are allowed to use ChatGPT as a helper tool, but you are not allowed to use it for plagiarism. Ensure that all work is original and properly credited.